Kernel similarity and frozen block reuse in CNNs

This study compares three independently trained plain CNNs and three residual CNNs on CIFAR-10. Local kernel resemblance and whole-block substitutability answer different questions. No plain CNN non-self replacement lost less than one validation percentage point. A small absolute drop and a positive advantage over removing the recipient must be distinguished.

Complete coverage is 12 structural snapshots, 132 original/control variants, 396 metric sets, 1,980 cutoff results and 4,800 scientific interventions across the two image splits. Imported origins and newly computed stages distinguish retained measurements from local production. All six completed baselines were reused for this release.

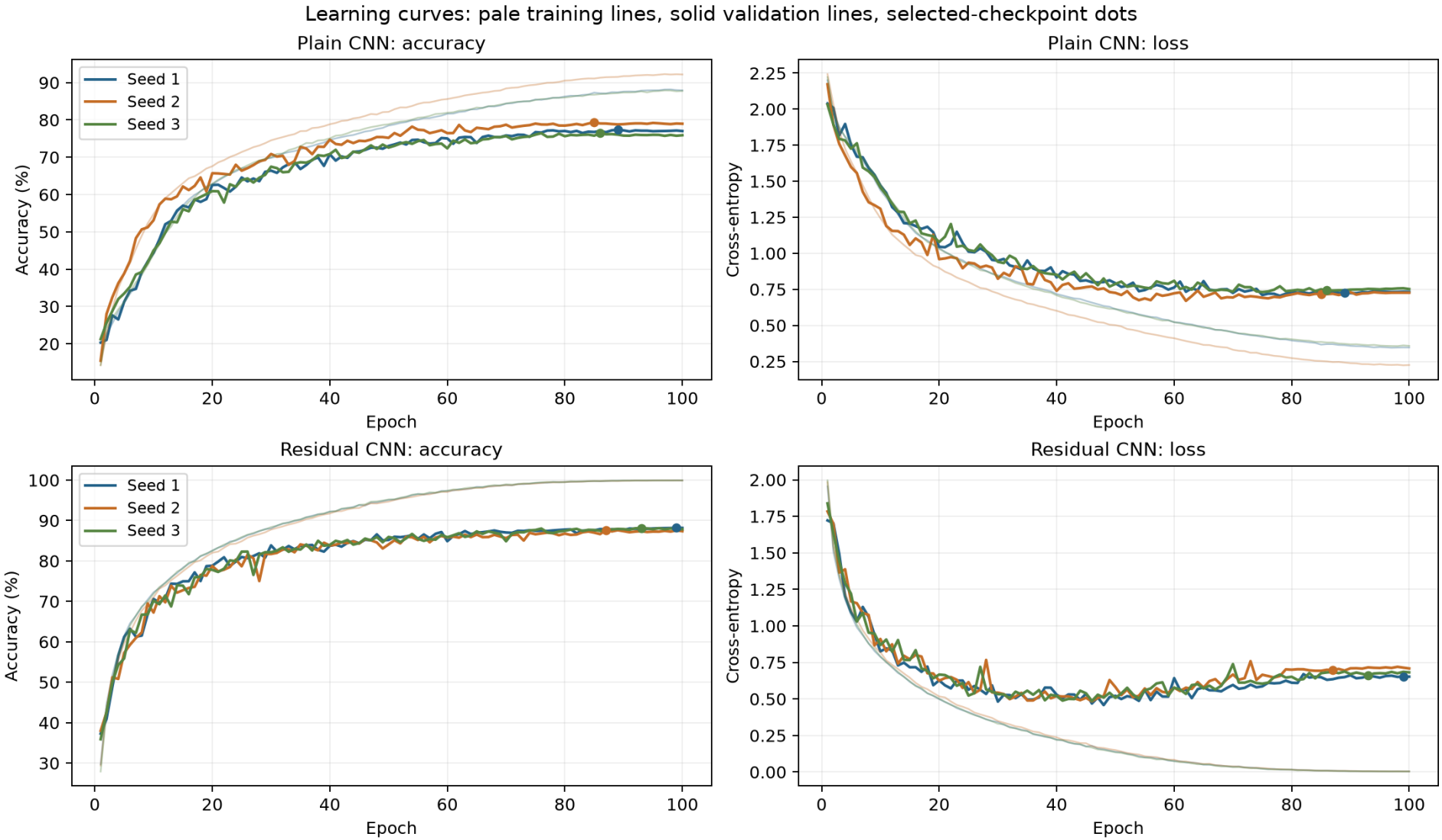
Models and learning

All models have 20 independently weighted blocks, two convolutions per block, width 64 and GroupNorm. The residual model adds the input to the learned branch before its final ReLU. Each run completes 100 epochs. The checkpoint with best validation accuracy is selected, with loss breaking ties. Validation uses 5,000 fixed held-out images; diagnostics use 1,000 fixed training images. The official test set is not used.

Model	Best epoch	Validation accuracy	Validation loss	Diagnostic accuracy
Plain seed 1	89	77.40%	0.725	85.50%
Plain seed 2	85	79.32%	0.717	89.20%
Plain seed 3	86	76.48%	0.743	84.70%
Residual seed 1	99	88.22%	0.652	99.10%
Residual seed 2	87	87.66%	0.697	99.10%
Residual seed 3	93	88.14%	0.661	99.20%

Achieved accuracy and training behavior matter when comparing architectures. Preserving poor baseline performance would not establish useful reuse. These models compare the outcomes obtained under the recorded protocol; they do not isolate an architecture-only cause.

Training and validation curves



All models complete 100 epochs. Training uses augmented changing batches; validation uses fixed images. Dots mark the accuracy-selected checkpoints.

Local weight similarity

Every eligible signed 3 by 3 input-output kernel is compared within its own checkpoint. Shape distance normalizes each nine-weight vector. Weight-value distance divides raw Euclidean difference by the smaller norm. Centered shape subtracts the kernel mean before normalizing. Sign reversal, rotation and reflection remain different patterns. The main cutoff is 0.05; for normalized shapes it means cosine similarity at least 0.99875.

Model	Shape pairs	Weight pairs	Centered pairs	Grouped shape kernels	Largest group / component
Plain seed 1	28	0	17	0.0329%	2 / 3
Plain seed 2	1,495	105	20	0.2859%	5 / 404
Plain seed 3	67	0	42	0.0744%	2 / 5
Residual seed 1	108	4	37	0.1280%	2 / 3
Residual seed 2	103	1	20	0.1207%	2 / 3
Residual seed 3	112	6	39	0.1341%	2 / 3

A tight group requires every member to match every other member. A graph component allows chains and can be much larger. A large connected component is therefore not a large family of near duplicates. Similar local slices also need not connect features with the same meaning.

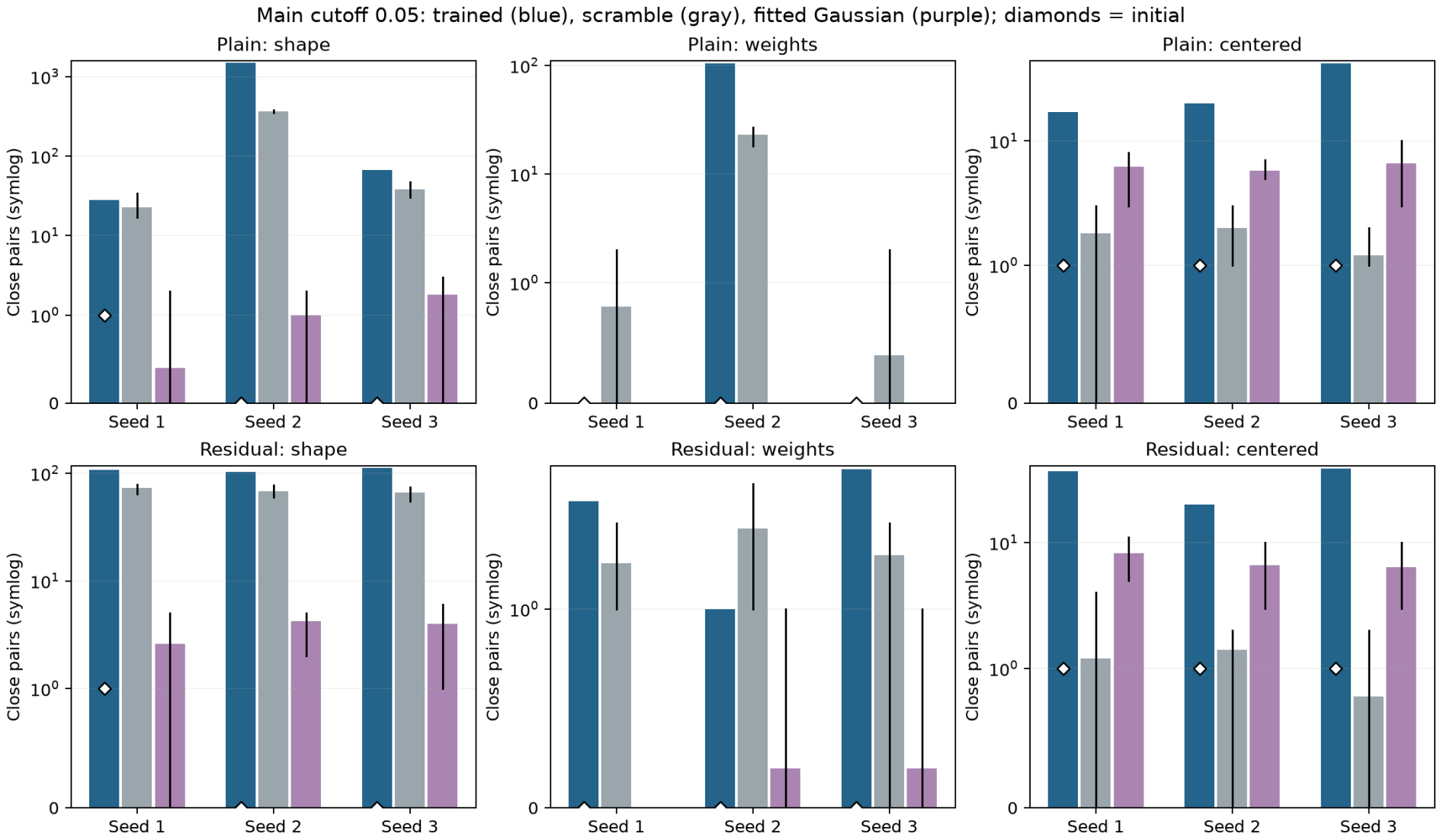
Controls and fixed cutoffs

Initialization provides the untrained reference. Spatial scrambling preserves each kernel's values, mean and strength while disrupting arrangement. A fitted Gaussian reference approximates layerwise shape concentration; normalization changes its realized moments, so it is not an exact learned-weight null. Control repetitions are not independently trained networks.

Model	Initial pairs	Scramble mean [range]	Fitted mean	Matched constant energy	Typical constant energy
Plain seed 1	1	22.8 [17-34]	0.4	97.54%	29.53%
Plain seed 2	0	364.6 [347-380]	1.0	98.92%	24.02%
Plain seed 3	0	38.0 [30-47]	1.8	97.02%	31.55%
Residual seed 1	1	73.0 [64-78]	2.6	95.81%	47.70%
Residual seed 2	0	67.8 [60-77]	4.2	96.06%	47.82%
Residual seed 3	0	66.4 [55-74]	4.0	95.46%	47.95%

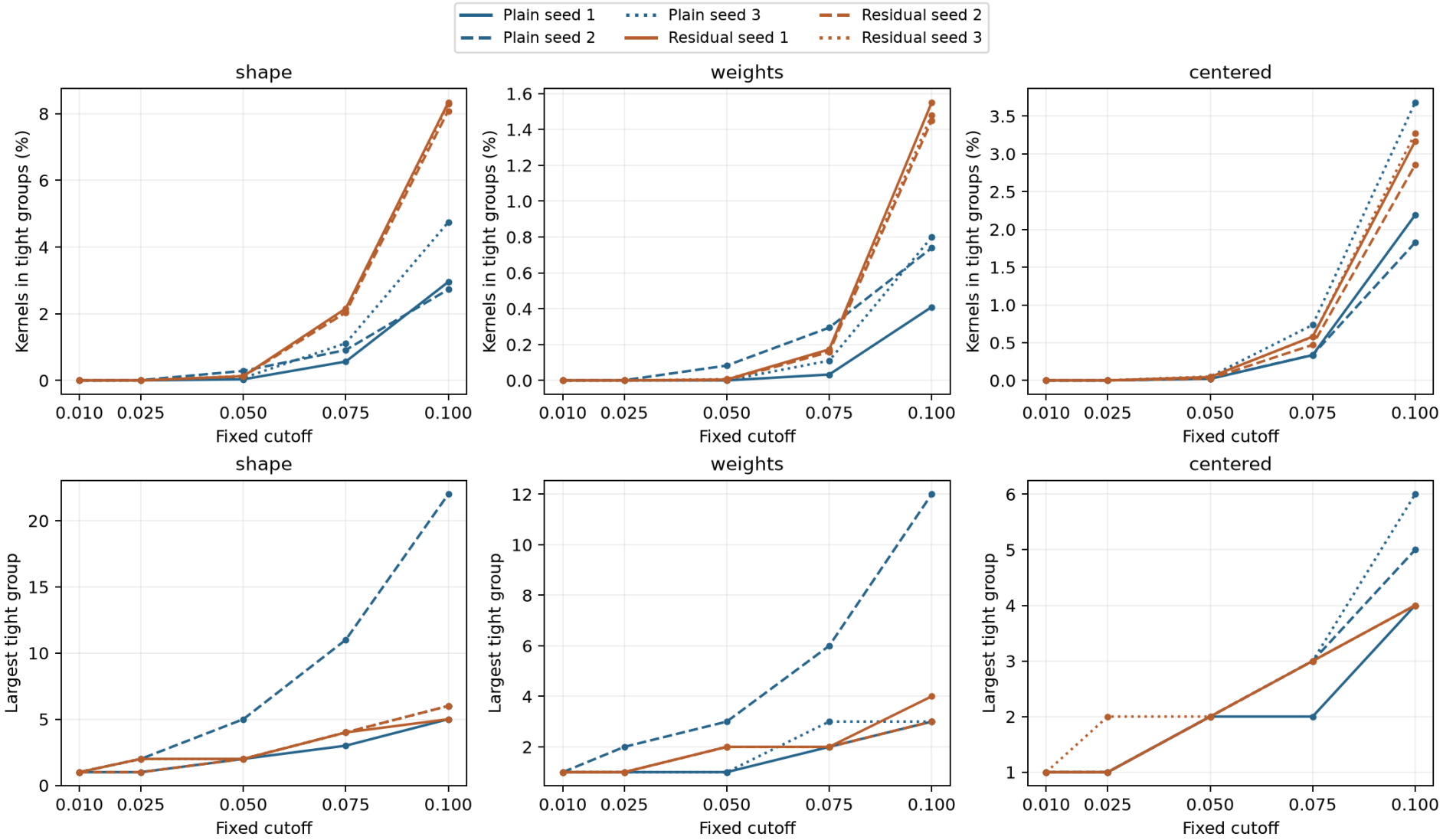
Constant energy is the fraction of squared weight strength explained by a uniform patch. High values identify nearly flat shapes. The medians compare kernels with at least one direct shape-match partner against all eligible kernels. A matched kernel can still be a singleton in the tight-group partition. Centered comparisons ask whether the nonconstant spatial details also match; they change the representation and need not find the same pairs. The numbers of shape pairs that also meet the centered criterion are Plain seed 1: 0; Plain seed 2: 0; Plain seed 3: 0; Residual seed 1: 1; Residual seed 2: 0; Residual seed 3: 2. These compare the exact pair identities, not just total counts.

Scientific controls at the main cutoff



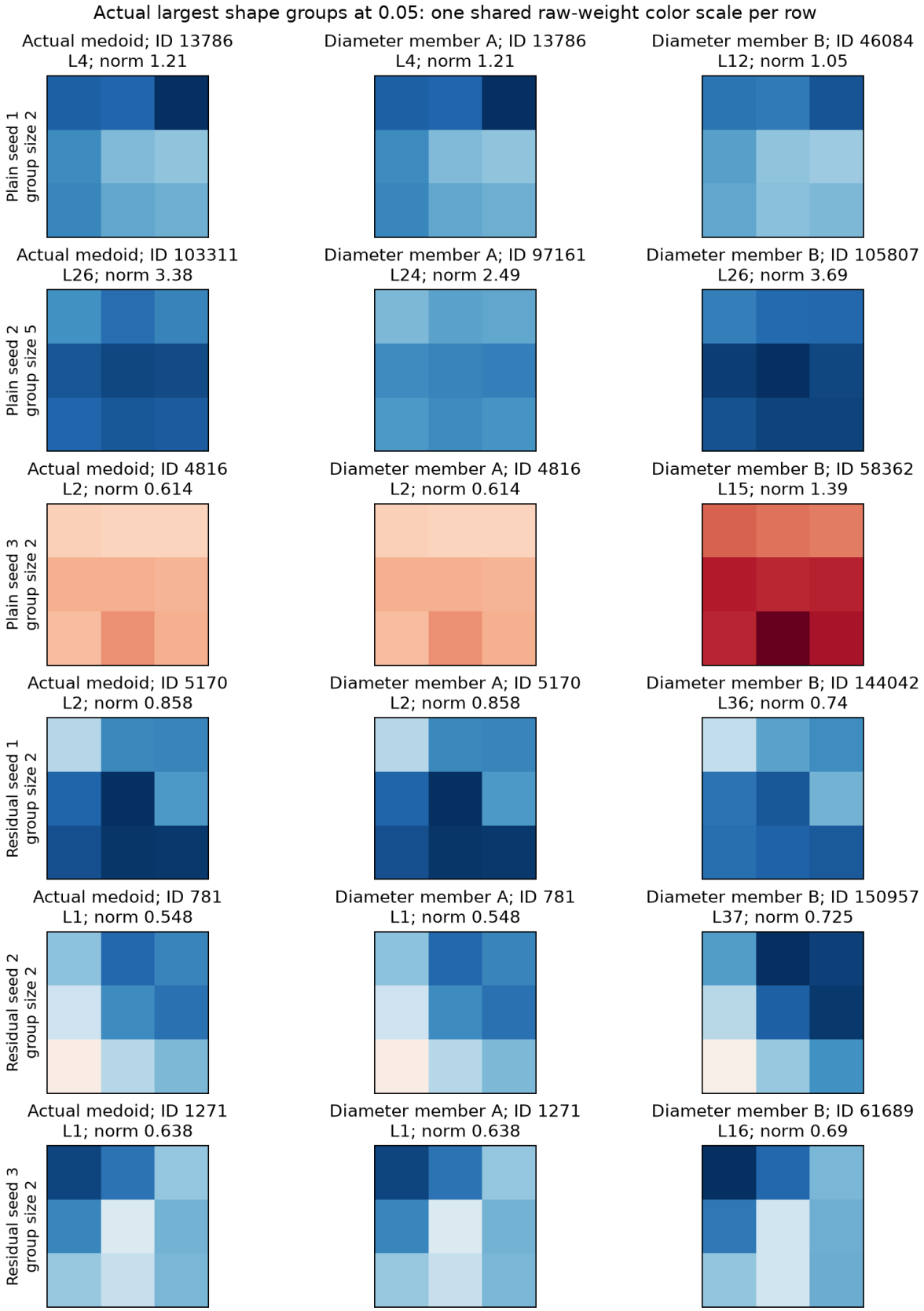
Whiskers show the observed five-control minimum and maximum, not uncertainty across trained models. The symlog axis keeps zero counts visible.

All three distance definitions across fixed cutoffs



Each line is one independently trained model. The cutoffs 0.01, 0.025, 0.05, 0.075 and 0.10 are fixed. Wider tolerances can add many edges while tight groups stay small. Read the full range: group membership requires every pair to meet the cutoff, while connected chains are a separate graph statistic.

Actual shape-matched kernels



A group medoid is an actual kernel; diameter members are its most separated pair. Shape agreement permits different strengths. A two-member group repeats one member in its medoid column.

Frozen block replacement and bypass

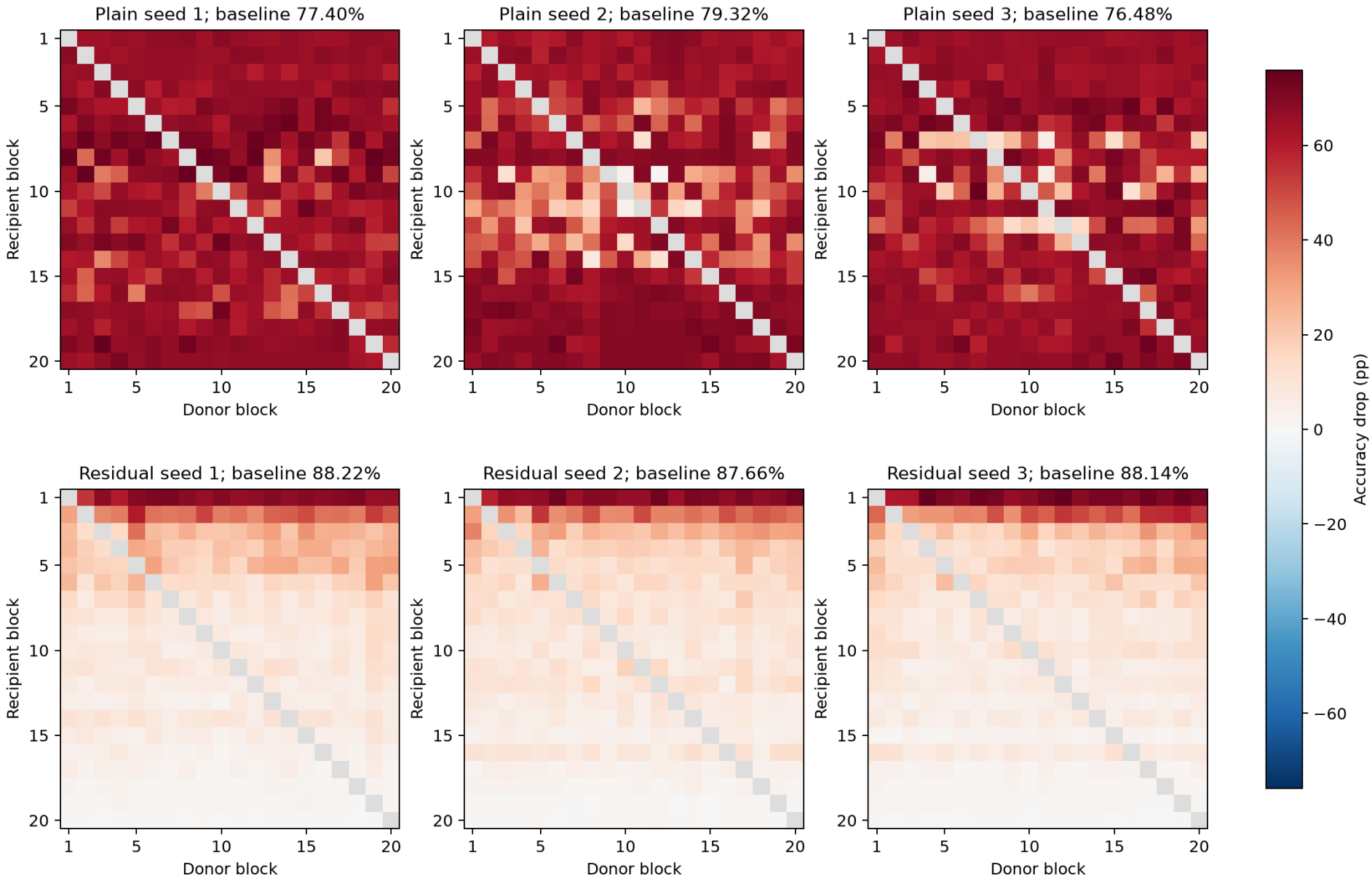
Each checkpoint is tested with all 380 ordered non-self replacements and 20 bypasses on both image splits. A donor's complete frozen block is copied into one recipient; the donor remains at its own position. Original weights are restored between trials. There is no retraining, adaptation or cutoff tuning.

Accuracy drop is reference minus modified accuracy in percentage points (pp); negative values indicate improvement. Loss increase is the change in cross-entropy, disagreement is the fraction of changed class predictions, and KL is reference-to-modified probability divergence in nats. Each column below is the median of its own metric.

Model and intervention	Drop pp	Loss increase	Disagreement	KL nats
Plain seed 1 replacement	66.18	4.829	88.82%	5.160
Plain seed 1 bypass	65.20	4.491	88.26%	4.802
Plain seed 2 replacement	64.44	4.450	84.67%	4.778
Plain seed 2 bypass	62.44	4.160	82.74%	4.521
Plain seed 3 replacement	65.35	4.501	88.76%	4.714
Plain seed 3 bypass	63.10	3.869	85.88%	4.040
Residual seed 1 replacement	7.46	0.526	15.02%	0.614
Residual seed 1 bypass	4.06	0.227	10.74%	0.322
Residual seed 2 replacement	8.92	0.582	16.65%	0.693
Residual seed 2 bypass	6.20	0.338	13.28%	0.450
Residual seed 3 replacement	7.29	0.513	15.09%	0.634
Residual seed 3 bypass	4.03	0.251	11.19%	0.367

The directional criterion is drop strictly below 1 pp, including improvements. Absolute change below 1 pp is recorded separately. The inherited one-point criterion is descriptive, not an equivalence test. Plain qualifying counts are 0 / 0 / 0; residual counts are 13 / 17 / 22 out of 380 each. Requiring both a drop below 1 pp and an advantage over bypass leaves 0 / 0 / 0 plain pairs and 3 / 10 / 5 residual pairs.

Validation replacement accuracy damage



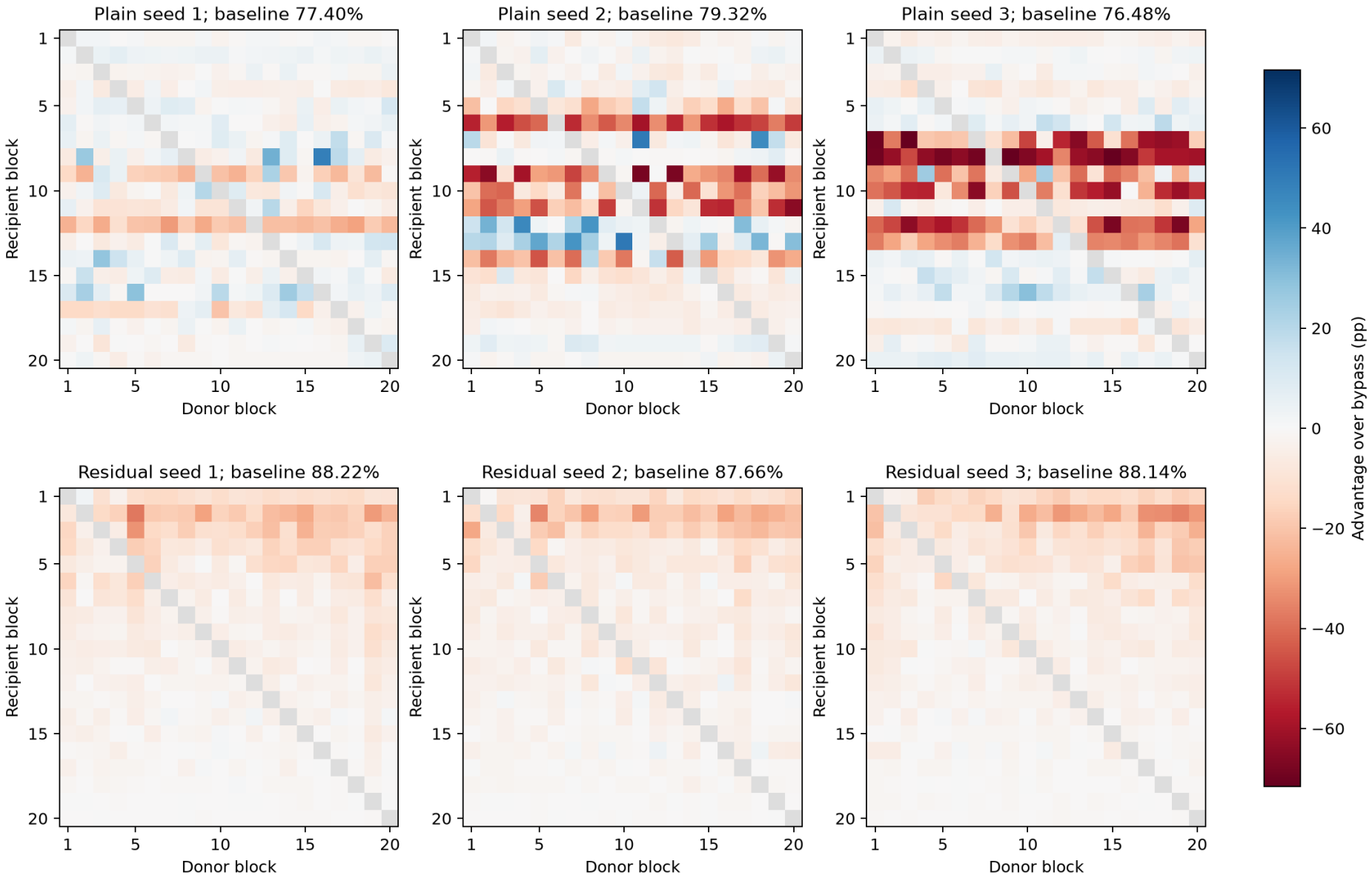
Rows are recipients; columns are donors. Positive values mean damage. All six panels share a scale. Gray cells are excluded self pairs.

Advantage over bypass is bypass drop minus replacement drop. Positive means the substitute works better than removing the recipient. It does not mean the substitute preserves the original network's accuracy or behavior.

Model	One best accuracy pair	Accuracy	Drop pp	Bypass advantage pp	Disagreement
Plain seed 1	B08 <- B16	56.82%	20.58	48.66	37.84%
Plain seed 2	B09 <- B12	78.18%	1.14	0.50	6.30%
Plain seed 3	B09 <- B04	72.10%	4.38	25.50	15.10%
Residual seed 1	B20 <- B08	88.02%	0.20	0.28	3.14%
Residual seed 2	B20 <- B06	87.26%	0.40	0.42	3.32%
Residual seed 3	B20 <- B04	88.00%	0.14	-0.16	3.86%

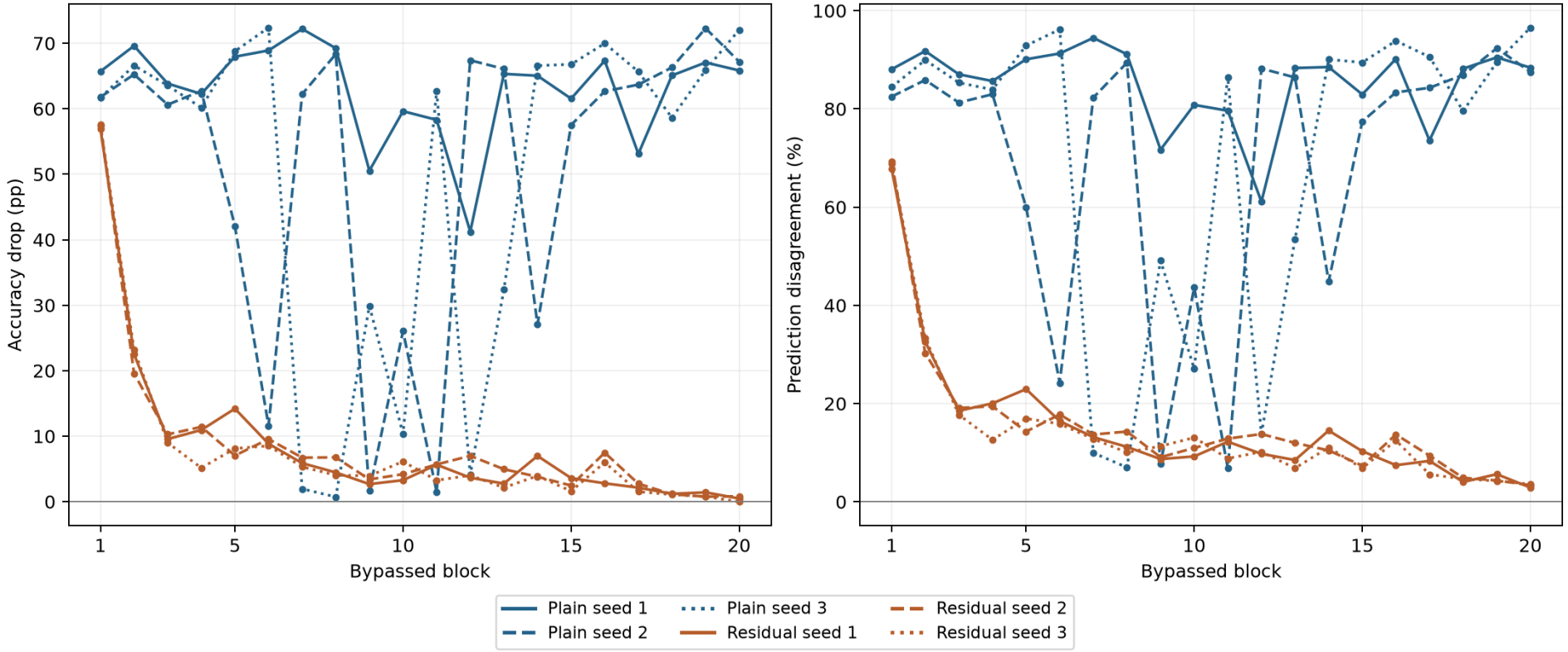
Recipient <- donor labels are one-based. If accuracy ties, the smallest recipient and donor labels are displayed; all measured pairs remain in the table. A small net accuracy change can hide offsetting lost and gained correct answers, so disagreement and probability metrics remain necessary.

Replacement advantage relative to bypass



Positive values favor the donor over removing the recipient. A replacement can beat bypass while still losing substantial absolute accuracy; read together with the damage figure.

Effects of removing individual blocks



Removing a complete block measures its influence under this intervention. Low removal cost does not identify another block performing the same computation.

Replication and limits

Plain seed-pair damage-rank correlations are 1-2: 0.029, 1-3: 0.056, 2-3: 0.065. Main-cutoff internal-layer shape-frequency correlations are 1-2: -0.060, 1-3: -0.017, 2-3: -0.005.

For pairs with drop below 1 pp and advantage over bypass, the shared and union counts are 1-2: 0 shared, 0 union; 1-3: 0 shared, 0 union; 2-3: 0 shared, 0 union. Shared means qualifying in both seeds; union means qualifying in either seed.

Residual seed-pair damage-rank correlations are 1-2: 0.863, 1-3: 0.882, 2-3: 0.887. Main-cutoff internal-layer shape-frequency correlations are 1-2: 0.031, 1-3: 0.126, 2-3: 0.017.

For pairs with drop below 1 pp and advantage over bypass, the shared and union counts are 1-2: 0 shared, 13 union; 1-3: 0 shared, 8 union; 2-3: 0 shared, 15 union. Shared means qualifying in both seeds; union means qualifying in either seed.

Diagnostic-to-validation damage correlations within each trained model are plain seed1: 0.984; plain seed2: 0.991; plain seed3: 0.985; residual seed1: 0.995; residual seed2: 0.995; residual seed3: 0.994. Those image splits do not add independent trained replications. Empty success-set overlap is undefined, not zero.

Three training seeds in each of two architectural families give six trained models. Kernel pairs, control repeats and 380 correlated interventions are not extra models. Position-aligned correlations do not identify equivalent features, and local similarity does not prove an interchangeable block or computational loop. These frozen interventions do not test trained sharing, adaptation after sharing, or recurrence in other architectures.

Local numerical results

Baseline summaries, all interventions, functional summaries, all structural results, kernel controls and overlap, layer profiles, depth profiles, functional stability and structural stability retain full precision. Accuracy and disagreement are fractions; drops and advantages are pp; constant energy is a fraction. Undefined quantities are blank in CSV and null in JSON. analysis.json contains the same canonical results. Primary functional metrics are calculated from project-local saved logits in float64; imported runner values remain unchanged under runs/. No model is evaluated while producing this report.

Kernel explorers expose all original snapshots, distances, cutoffs and coordinates, with full-precision CSV export.